

DEN161 Honors Assignment

Ian Dodson

Background: Why Inclusive Design Matters

In the world of engineering, it is extremely important to be inclusive. Inclusiveness in engineering is all about making your product universally functional. This can include everything from “age to culture, economic situation, gender, language, and human factors” (Full Stack Academy, 2024). Anything that prevents someone from using your product could be seen as a negative in the design process, so an engineer’s goal is to make a product that not only functions properly but can be used by as many people as possible. This is a very hard thing to do but it can be the difference between an okay product and a universally used, world-changing one. When designing for inclusiveness, an engineer needs to look at not only who is going to be using their design, but who might want to use their design; this is where it gets tricky. An example of this issue is seen with computer and gaming accessories. These items have been carefully designed and iterated for several decades to feel as comfortable to hold and use as possible. However, they have been designed around the average person. This causes a major issue for people with deformities or disabilities that make these designs unusable for them.

One company that has been successful in tackling this issue is Microsoft with their Adaptive Accessories. These are products made to help people customize their device accessories to computers easier to use. Their system consists of three parts: the “Adaptive Mouse, Adaptive Hub, and Adaptive Buttons” (Microsoft, 2025). This product has an amazing story behind it and reflects how special and important inclusive designs can be. The Microsoft Adaptive Accessories were designed by Microsoft’s Senior Designer, John Helmes, for his daughter who suffers from cerebral palsy. Her condition affects her motor skills and makes using accessories like mice and controllers difficult. While John’s design was originally only intended for his daughter, the world changing application became apparent and Microsoft began to produce these products for people around the world. They also expanded their designs to more than just motor skills and can now help people with all kinds of disabilities use computers and gaming systems effectively!

Another successful inclusive design is IKEA with their ThisAbles Project. IKEA noticed that people with special needs struggled to use their products as their standard furniture often included functionalities that are hard to understand or require fine motor skills. To combat this issue, IKEA started an initiative to help “people with special needs use standard furniture through smart additions.” (VIJ, 2025). ThisAbles uses smart, inclusive designed additions to their products to make them universally accessible. These additions include glass bumpers, snap on clips, curtain grippers, and friendly zippers.

While inclusive design is an amazing development in the modern day, sometimes they backfire and have a negative effect. In engineering, often times products are not given enough time to be fully fleshed out and complete. This can cause products to not function as intended. When a product doesn't work as intended the public can feel hurt, the company who made it will look bad and the engineer who designed it can receive major criticism. Another contributing factor to this issue is that the world is changing so much, it can be hard to account for all the complexity and nuance in society. This is not so much for physical disabilities, but the world has a lot of social constructs and opinions that are really hard to design around. This is just another factor that goes into the design process, and it is important as an engineer to be aware of the way your design is going to be perceived once it is out in public.

One example of a negative effect as a result of inclusive design is through text to speech. Text to speech is an important product that not only is used for convenience when someone's hands are full, but also as a necessity as "individuals who cannot type due to disabilities such as dyslexia, blindness, hand tremors, or arm injury" rely on it to send messages (Candidate X, 2025). This tool, however, has returned some alarming patterns. On average, the error rates for text to speech are almost twice as frequent for black users as it is for white users. This shows how this product, while trying to be an inclusive way for people to text, actually discriminates against people. This is the opposite of what this product is meant to do, and it is extremely sad and harmful to see.

Another example of a negative effect from a poor design is with airport body scanners. Airport body scanners are designed to allow all types of people to go through the scanner and find abnormalities in hope of keeping air travel safe for all people, "however these scanners tend to discriminate, particularly against transgender individuals, Black women, Sikhs, and Muslims" (Iyamah, 2025). Once again, a product that is meant to be used by all types of people is discriminating against whole groups of people and hurting them. When these scanners come up with errors against these groups, it creates an uncomfortable and dangerous situation. People are searched and exposed for no reason. This can be extremely traumatic and painful and inexcusable. This is why it is so important to do inclusive design right.

DEN161 Project Summary

Our project in DEN 161 was to design and build a gaming controller and write a project report to go along with it. This project challenged us to think like an engineer as we go through all of the steps that an engineer would go through if they were to actually design a controller for production. We were given criteria and constraints that we had to meet in order for our design to be successful. These included functionality, comfort, size, safety, and cost. This class and project were extremely beneficial to me as it was my first time building something from scratch that worked and I was super proud of my team!

Some of the risks of our design are aligned with a lot of the risks of modern-day controllers. It was designed around the average person. This means that it does not accommodate people with missing fingers, poor motor functions, or cognitive disabilities. Some of the components were even difficult for the “average-person” to use. The joystick controls the mouse which, while functional, makes it hard to use as instead of having all the muscles in your arm to aim the cursor, you only have the use of your thumb, and is thus hard to aim and adjust. People who struggle to control their muscle strength would have a significant challenge if they tried to use the joystick on our controller and could be discouraged by it. Our controller was also not the most comfortable. There was no case around it and the thin PCB board was used as the body. This made it sit oddly in your hand and a little pointy. The buttons and joysticks were also put too close together and could make it uncomfortable for users with larger hands to use.

Design is the most important factor in making a good, inclusive product. When we started the semester, we didn't have any plans for what we wanted our controller to look like, but as we learned more about the engineering design process, we started to implement more and more components and made sure to fit withing the requirements. If we were to design a new product now with all of the knowledge from the course, there would be a lot of things we would do differently that would have made our design more inclusive and better overall. Now that I have a better understanding of what design is and why it is important, I will apply what I know about inclusive design and product manufacturing into my future as an engineer and continue to learn how to make better, more inclusive products for everybody.

Project Recommendations

Some changes that could be made to our project include adding an outside shell, remapping the button layout, and changing the joystick functionality. By creating a shell for the controller to fit into, the controller will feel significantly more comfortable using. This is seen in all modern-day controllers. I've watched a lot of videos about building custom controller cases and have seen what the inside of the controller looks like. It is often a plastic case with a soldered board on the inside. At the base level, most controllers consist of the same components as our controller. That means this should be a relatively simple addition to implement into our design. All we have to do is design a plastic case for our board to fit in. By adding a case, our controller would be a lot more comfortable to hold and thus make it easier to use. This makes that design a lot more inclusive and improves it significantly. By changing where the buttons are laid out on the controller, the controller can be a lot easier to use. Right now, the joystick and buttons are a little too close together. This creates an unnatural resting position for your thumbs. This can cause cramping when the controller is used for a long period of time. I would also reangle the bumpers and move them slightly farther out or pick different bumpers entirely. The current bumpers are long and tiny, making them hard to use. By getting shorter and thicker bumpers, they are more easily accessible and more comfortable to use. I

would lastly make the joystick optional and create a mechanism to allow for customization. Instead of the joystick, you could use an added d-pad. This allows a user to suit the controller to their needs and makes it a better and more inclusive product.

These recommendations allow us not only to meet the criteria of the project but exceed them. While our original design satisfies the criteria we could always do more. This is seen in the real world of engineering as real world products go through a long process of designing, prototyping, and redesigning to create new iterations and perfect the product. We can implement that into our design by going further with this project and adding these new elements to our controller. The more we add the better it gets and the better we get at designing and engineering.

These additions will affect the design however, they will increase the cost, size, and feasibility. The more you add the more expensive it is going to be. This is because every new element comes at a cost. Another issue that affects the cost is creating custom parts. When you create custom parts, the manufacturing cost goes way up as it is more inconvenient to create one part than it is to mass produce something. By adding these parts and the shell, the size of the controller will increase which could jeopardize the size constraint of the project. It is imperative that it stays within the limit and below the cost established by the project as in this exercise they are set by a client ordering your product. By changing the layout of the controller, it shouldn't change the cost, but it will be hard to design so it is important to be smart and patient with how you implement all of the additions.

I have learned a lot from this class and the controller project. It has taught me all about the design process and what it looks like to be an engineer. Over the course of this semester, I made a lot of mistakes, but I have learned from them, and it has made me better. I will continue to make mistakes and learn from them over my next couple of years at K-State and throughout my career as an engineer. I can wait to go out into the real world and apply what I have learned at school to the real world!

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